

PDF TO AUDIO BOOK CONVERTER

A Mini Project submitted in partial fulfillment of
the requirements for the degree of

Master of Science in Computer Science

to the

Periyar University,Salem-11

By

G .MENAGA

Reg No:C21PG185CSCO05



ADHIYAMAN ARTS AND SCIENCE COLLEGE FOR WOMEN,UTHANGARAI.

(Affiliated to Periyar University,Salem-11)

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Srinivasa Nagar, Uthangarai-635 207

2022

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UTHANGARAI.**

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2022

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Date:

CERTIFICATE

This is to certify that a Mini Project Work using Open Source entitled
“PDF TO AUDIO BOOK CONVERTER” submitted in partial fulfillment of
the requirements of the degree of **Master of Science in Computer Science** to
the Periyar University, Salem is a record of bonafide work carried out by
G.MEANAGA Reg. No C21PG185CSC005 under my supervision and
guidance.

HEAD OF THE DEPARTMENT

INTERNAL GUIDE

Date of Viva-voce

INTERNAL EXAMINAR

EXTERNAL EXAMINAR

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G.MENAGA

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ABSTRACT

PDF TO AUDIOBOOK CONVETER

ABSTRACT

In this project is the PDF to Audio Converter is proposed. It provides an alternative way to access the books and any pdf file for lazy, readers, and others. Using this PDF to Audio Converter the user will be able to listen to his/her favorite PDF and can do their daily routine.

The application can be used to read any PDF . The following application can be used to convert text from PDF to audio using Tkinter and python files, functions, and definitions.

Tkinter is the python binding to the Tk toolkit which is used across many programming languages for building the Graphical user interface which is a GUI.

The main packages used in this audiobook converter are pyttsx3 and PyPDF2 libraries. To overcome the(issues) the project PDF to Audio Converter has been developed to extract data from the pdf selected by the user, and to extract the data from the pdf, convert it to audio format to read out loud.

This is a python library built as a PDF toolkit. To overcome the(issues) the project PDF to Audio Converter has been developed to extract data from the pdf selected by the user, and to extract the data from the pdf, convert it to audio format to read out loud.

It will allow the user to note down important notes and it also has the feature of save, cut, copy, and paste. Index Terms -

[Python, pyttsx3 and PyPDF2, Text to Speech, Converter, Tkinter.](#)

INTRODUCTION

1.INTRODUCTION

Text-to-speech and related read audio tools are being widely implemented in an attempt to assist students' reading comprehension skills. PDF to the audio system is a screen reader application designed and constructed for an effective audio communication system.

PDFs were designed to present and exchange documents reliably, PDFs are an open standard document format used globally, maintained by the International Organization for Standardization (ISO).

The document format is one of the most convenient methods for electronic communication, and also for the exchange of information.

The PDF to the audio system will power text on screens to read aloud (speak) with support for many languages . The PDF to Audio Converter project provides an alternative to access the PDF books for the blind, lazy, readers, and others.

The following application can be used to convert text from PDF to audio using Python predefined libraries like pyttsx3 and PyPDF2 and also using some Tinker files.

In the current busy routine people do not have time to take a book and spend time reading it, instead, everyone needs alternative access to read the content. If a person is traveling, he\she cannot read a book, instead of reading, they can listen to it.

The PDF to Audio Converter project provides an alternative to access the PDF books for the blind, lazy, readers, and others. Using this PDF to Audio Converter the user will be able to listen to his\her favorite PDF and can do their daily routine.

1.1SYSTEM CONFIGURATION

1.1.1HARDWARE CONFIGURATION

MEMORY SPACE:

- MOUSE :Touch Pad
- INPUT :Keyboard,Mouse
- RAM :8GB
- PROCESSOR : 11th Gen Intel(R) i3-1115G4 @ 3.00GHz
- MINIMUM :32 MB
- RECOMMENDED :64 MB
- ETHERNET CONNECTION :wireless adapter(Wi-Fi)
- HARD DISK

1.1.2 SOFTWARE CONFIGURATION

- PYTHON :version 3.11
- OPERATING SYSTEM :windows 11
- CODING LANGUAGE :Python
- TOOL :IDLE,pycharm

1.2.3 SOFTWARE SPECIFICATION

PYTHON LANGUAGE

Python is a high-level, general-purpose programming language. In this python language having so many packages. It is dynamically-typed and garbage-collected.

Python was conceived in the late 1980s by Guido van Rossum at Centrum Wiskunde & Informatica (CWI) in the Netherlands as a successor to the ABC programming language, which was inspired by SETL, capable of exception handling and interfacing with the Amoeba operating system. Its implementation began in December 1989.

It interfaces to many OS system calls and libraries and is extensible to C or C++. Many large companies use the Python programming language include NASA, Google, youtube, etc.

The main characteristics of python

- Multi-paradigm programming language
- Interpreter language
- Interactive language

PYTHON FEATURES

1. Easy to learn - Python includes a small number of keywords, precise structure and well defined syntax.

2. Easy to read - python code is clearly defined and visible to the naked eye.

3. Easy to maintain - Python source code is easy to maintain.

4. Standard General Library - python's built library is very portable and shortcut compatible with UNIX, Windows, and Macintosh.

5. it can be used as scripting language or compiled into Byte-code for large-scale application development.

SYSTEM STUDY

2.SYSTEM STUDY:

Background Study:

On July 1, 2008, the ISO issued ISO 32000-1:2008 and disseminated it as an open standard. When it came to defining the term, the responsibility was transferred to an ISO Committee of industry experts. The Adobe Public Patent License, announced in 2008, grants ISO 32000-1 royalty-free use of all Adobe-owned patents necessary for PDF-compliant implementations to be produced, used, sold, and distributed. In 2008, this licence was made available to the public. As early as 1993, Adobe Systems released the PDF standard open source. Adobe's own PostScript format, as well as other formats like Common Ground Digital Paper and Farallon Replica were all opposing with PDF in the early years. As a starting point, the user provides a location for a PDF file to be read to them using audio, which is a simple way to association various Python modules.

2.1 EXISTING SYSTEM:

During the development of the research study, the following limitations were encountered;

1. Limited research material available at the school library and on the internet.
2. High cost of setting up the system as it requires a high programming language.
3. Combining school work and carrying out the research.

2.1.1 DRAWBACKS:

- To create a PDF to Audiobook Converter using different Tkinter and python files, functions, and definitions.
- The main packages used in this audiobook converter are pyttsx3 and PyPDF2.
- Pyttsx3 is a python library for text to speech. This is the reason which helps the machine to speak to us.

- This reads out the selected PDF into audio in a suitable voice, speed of reading with volume adjustments. This also contains a notepad.
- Pyttsx3 extracts the text from the PDF. This is a pure-python library built as a PDF toolkit .
- This can extract document information, split, and merge documents page by page.

2.1.2 DEFINITION OF TERMS

- **PDF** – Portable Document Format.
- **Audio** - Audio most commonly refers to sound, as it is transmitted in signal form.
- **Speech**- the expression of or the ability to express thoughts and feelings by articulate sounds.
- **Oral** - relating to the transmission of information or literature by word of mouth rather than in writing.
- **Reading Disabilities** - a condition in which a sufferer displays difficulty reading.
- **File** - a collection of data treated as a unit by a computer.
- **Document**- A document is a written, drawn, presented, or memorialized representation of thought.

2.2 FEATURES:

The basic feature in our project is to convert a .pdf file into a .mp3 file. For this, we have used two libraries (i.e. pyttsx3 and pyPDF2).

1. Pyttsx3 – It is a Python library for Text to Speech. It has many functions which will help the machine to communicate with us. It will help the machine to speak to us .
2. PyPDF2 – It will help with the text from the PDF and is capable of extracting document information, splitting documents page by page, merging documents page by page, etc.
 - First, the program will import the PyPDF2 and pyttsx3 modules.

- Then to open the Pdf file we have used PdfFileReader().
 - To extract the text from the page we have used extract text.
 - Then by using pyttsx3 it will convert text to speech
 - Then to select the page to be read we used the get Page() method.
 - Then from the pyttsx3 object, We used the say() and run wait() methods to speak out the text.
- We have added a few extra features to make our project more user-friendly like voice changer, speed, volume, and a notepad.

OTHERS

- ❖ Minimum time required.
- ❖ It is easy to read.
- ❖ It is easy to understand.
- ❖ It is also using to reading practice.
- ❖ Pdf to audio book converter is using all the places it can be access.

2.3 PROPOSED SYSTEM:

In this current busy routine people do not find time to read a book, or to convert the PDF file into an MP3 player using third-party applications or web applications.

Even I have a directory at which I store pdf books that I plan on reading, but I never do.

So, I thought hey, why do not I make them audiobooks and listen to them while I do something else !

In this system, we are developing a GUI application using python to convert the PDF file into audio format and read it out to the user

The application is more user friendly as it does not require any audio file or MP3 player. The user will have to select the PDF file which the user wants to listen to.

2.4 PROJECT DESCRIPTION:

In the current generation students, researchers, authors don't find time to read a book on an electronic device as that might strain their eyes and might face we have designed an application that extracts the text from the selected PDF and reads it out to the user.

Best free PDF to Audio converter software and online tools

Here is the list of tools added in convertinPDF to audio:

1. Balabolka
2. Online-Convert
3. Zamzar
4. AnyPDF.

1.BALABOLKA:

Balabolka is a free portable tool that lets you convert PDF to audio files. It lets you convert PDF to MP3, WAV, WMA, OGG, MP4, M4A, M4B, OPUS, and AWB format audio files. One of its unique features is that you can use its split and convert feature to convert a PDF file by size of text blocks, lines where all letters are capital, by named bookmarks, etc.

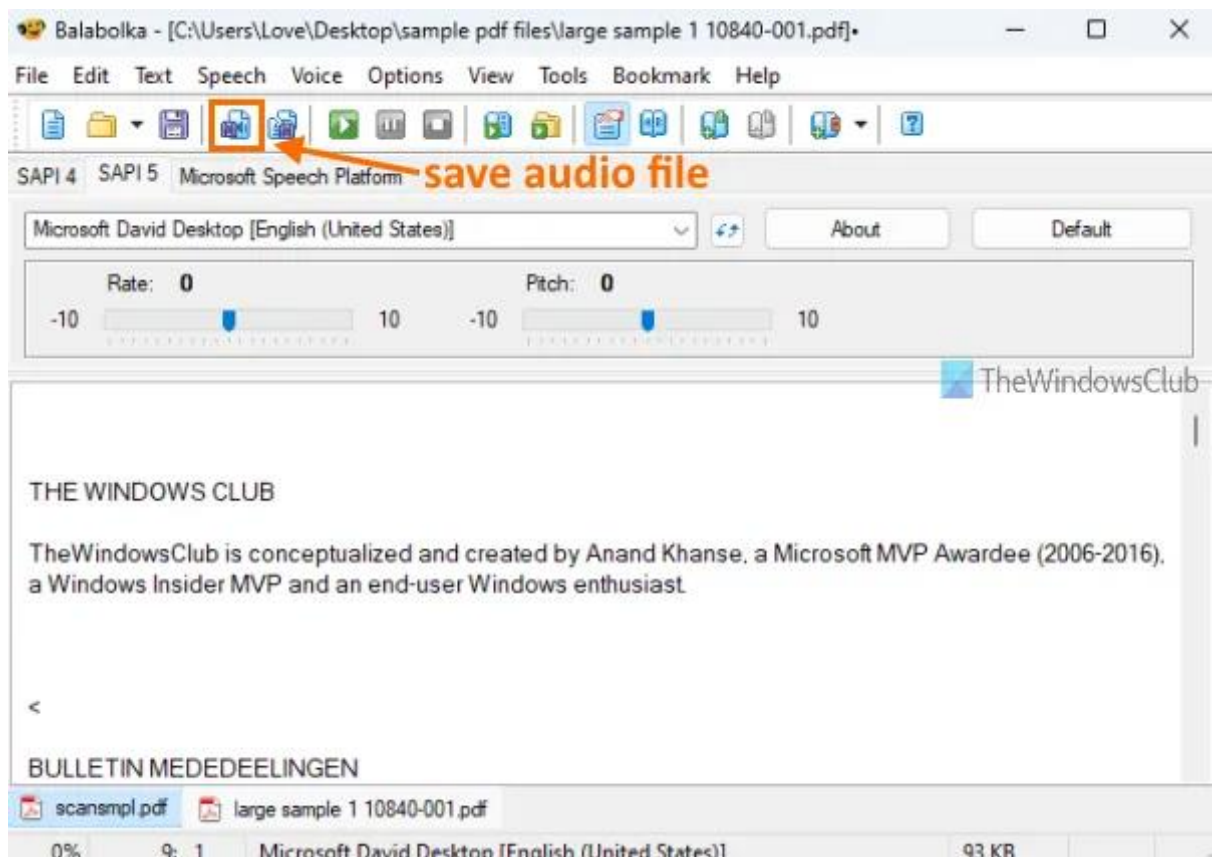
This tool also lets you open multiple PDFs in different tabs and then you can listen to its text content or simply create the audio files of those PDFs. In addition to that, you can also change the speech type (3 different speeches are available) and set the pitch, rate, and volume of the selected speech.

Apart from converting PDF to an audio file, this tool can be used to view and read the text from ODP, Markdown file, TXT, RTF, DOCX, DOC, XLSX, DJVU, XLS, EPUB, FB2, and other supported format files. So the tool is feature-rich and works quite well.

Now let's check how to use this tool for converting PDF to audio. Once you have downloaded this tool and opened a PDF file on its interface, follow these steps:

1. Click on the SAPI 5 tab available on the top left part of its interface

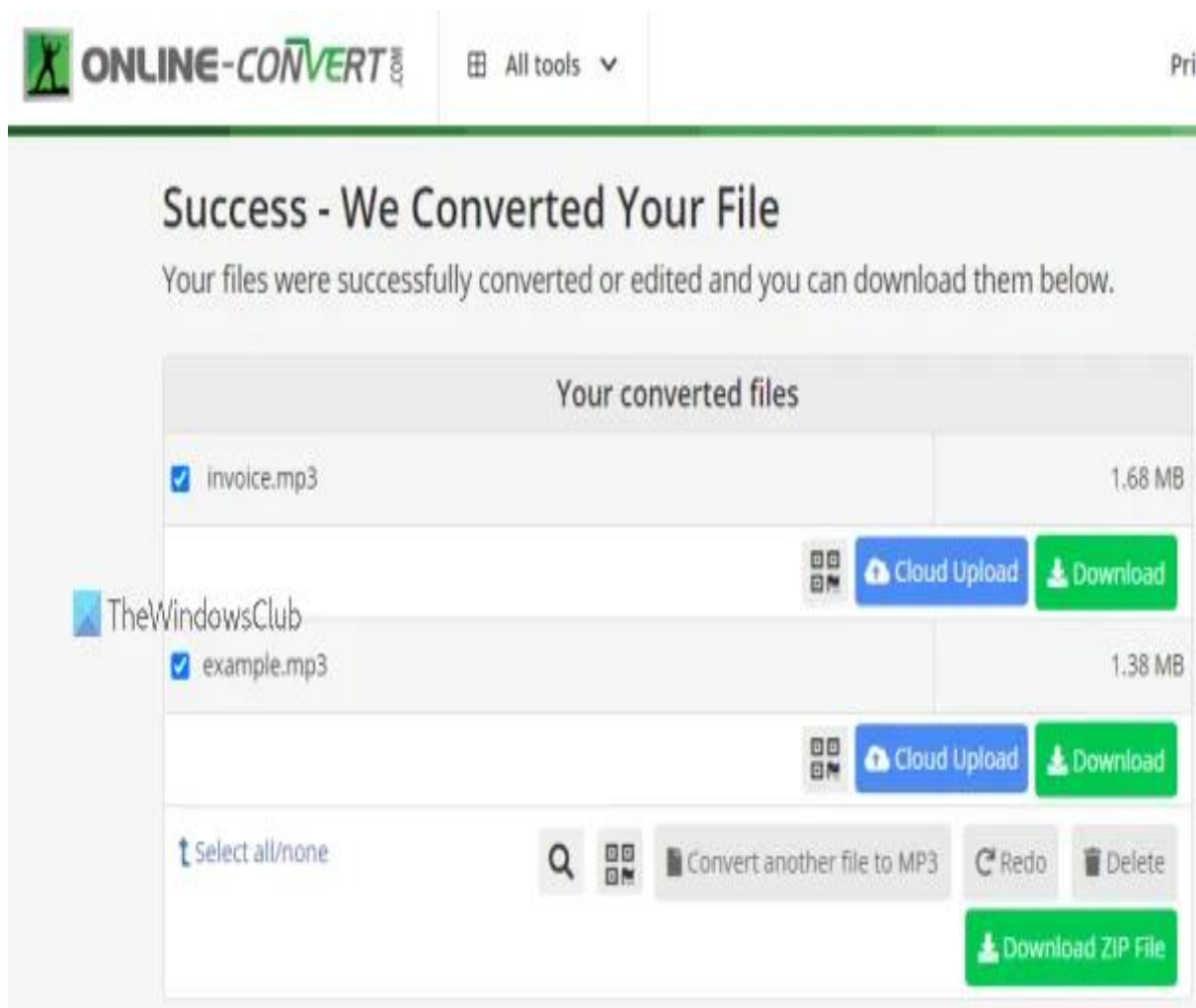
2. Select a speech from the drop-down menu
3. Set the pitch, volume, and rate
4. Listen to the text content so that you can find out if the audio is good enough for you
5. Click on the **Save Audio File** button present just below the Speech menu
6. A Save Audio File window will pop-up
7. There, select the audio format
8. Give a name to the audio file.



2)Online-Convert

Online-Convert is an all-in-one service for different conversion types. There are multiple conversion types such as archive converter, eBook converter, video converter, image converter, etc. A PDF converter tool is also present there using which you can convert PDF to MP3 files.

It's free registered plan, you can upload a maximum of three PDF files per conversion. However, there are limitations to the free plan. You can upload a maximum of **100 MB** PDF file and you **can convert only 3 PDF files in 24 hours**. If this limitation doesn't bother you, then this service is good to use.



3)Zamzar

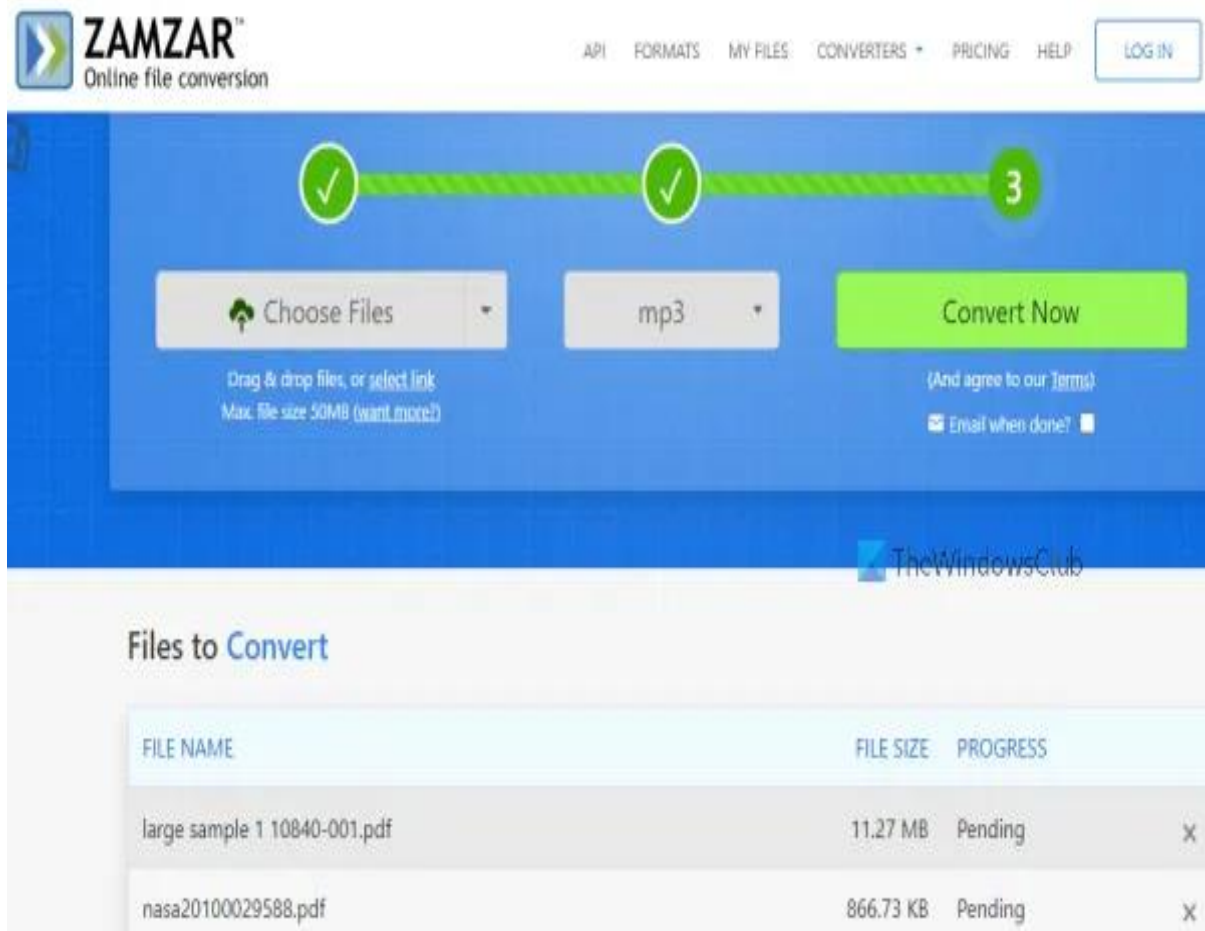
Zamzar service is similar to the Online-Convert service as it also provides a lot of conversion types. It has a document converter, audio video converter tools, eBook converter, archive converter, etc. Hundreds of file formats are supported for conversion. And, PDF to MP3 conversion is also possible here.

The features are pretty good but its free plan is very limited. In its free plan, you can upload a maximum of **50 MB** PDF file (less than 50 MB for 2 PDF files) and only **2 conversions** are supported per day.

For converting PDF to MP3, open its homepage. After that, click on **Choose Files** button to add PDF files from your computer. You can also use the drop-down icon available just next to Choose Files button to add a PDF document from your OneDrive, Dropbox, Box account, and Google Drive account.

Once the PDF file is added, click on the Convert To drop-down menu, and select the mp3 option available in the Audio Formats section. After this, press the **Convert Now** button.

Finally, the service will do the rest of your work. It will upload the added PDF file(s), start the conversion process, and provide you with the download links to save the output audio file. This website doesn't provide the options to set pitch, volume, or speech, but the output comes well.

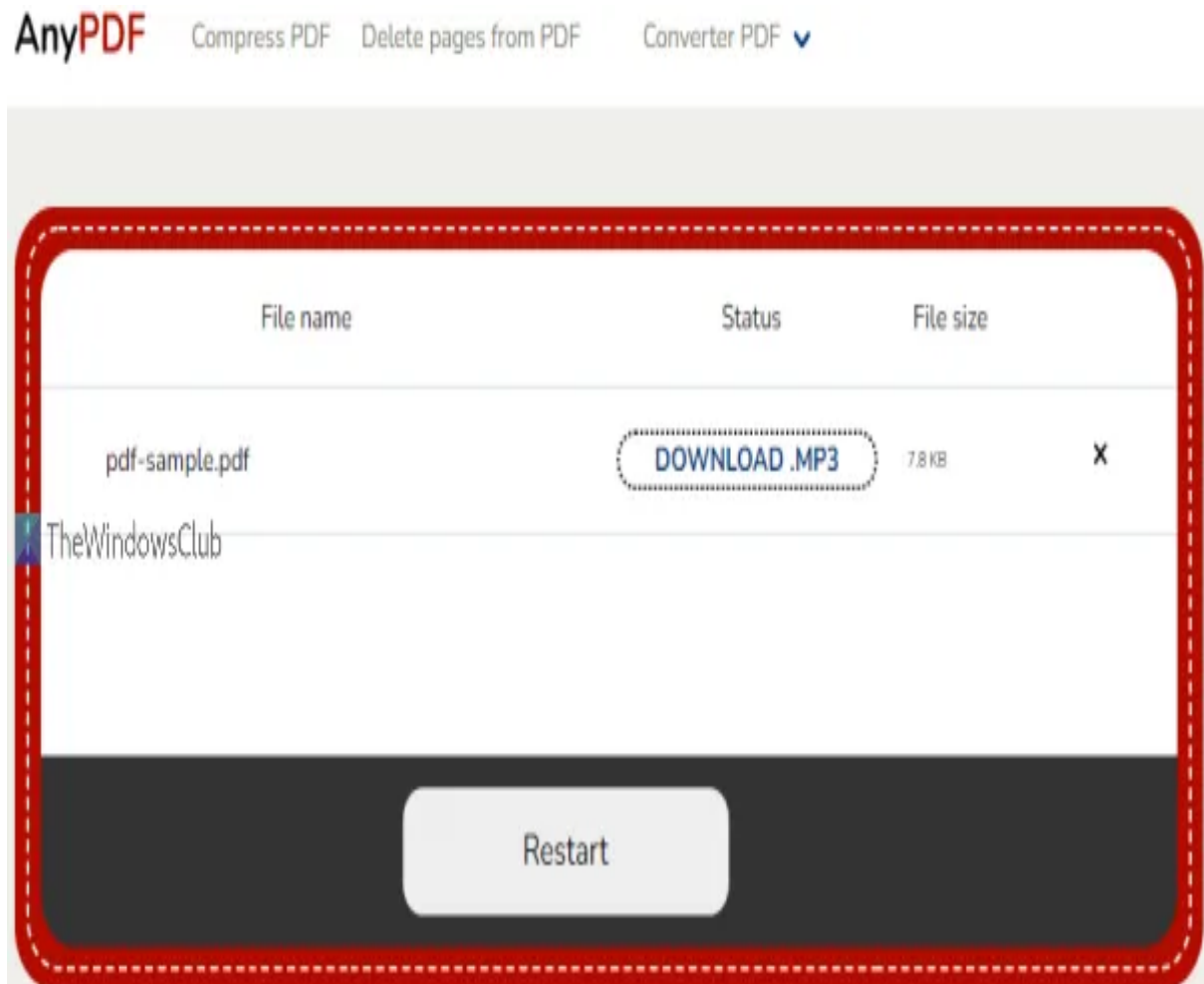


FILE NAME	FILE SIZE	PROGRESS	
large sample 1 10840-001.pdf	11.27 MB	Pending	x
nasa20100029588.pdf	866.73 KB	Pending	x

4. AnyPDF

Any PDF brings a very simple PDF to MP3 converter tool. It doesn't mention any size limit and the number of conversions per day. However, based on my testing, I found that the tool is good for basic PDFs having only the text content. While other tools covered in this post were able to convert other PDF files that I used for testing, it showed errors in processing such PDFs.

To use this PDF to MP3 converter tool, you need to access any-pdf.com service. After that, click on the **Choose files** button to add a PDF file from your desktop. Once done, hit the **Convert** button. When the file is processed, click on the DOWNLOAD button to store the output MP3 on your computer.



Steps to Build Python Pdf to Audio Converter Project

Below are the list of steps to convert Pdf text to audio speech and audio to pdf file using python:

1. Import Modules
2. Make constructor of the Application class
3. Functions to draw main frame
4. Function deleting frame
5. Speech to Text function
6. Text to speech function
7. Conversion functions
8. Function for reading and clearing files.
9. Main function

Modules

- os – This module interacts with the operating system.
- Tkinter module – Tkinter is the standard interface in python which is used for creating a graphical user interface.
- from tkinter import messagebox – Import message box separately for showing messages on the screen.
- speech_recognition – This library converts speech into audio file.
- win32com.client – win 32com.client module provides access to automation objects.

SYSTEM DEVELOPMENT

3.SYSTEM DEVELOPMENT:

Definition

The objective of this project is to create a GUI based PDF text to Audio and Audio to PDF converter. To build this, you will need intermediate understanding of Tkinter, OS, path, Speech Recognition and PyPDF4 libraries and basic understanding of the Pydub and pyttsx3 libraries and message box module.

pdf_to_audio()

This method, where we will convert text from a page of a PDF file to speech, our GUI window will be a Top Level widget, with two entry fields: the first one asking for the path to the PDF file, and the second one asking for the page number from which the text should be read.

speak_text()

The button in the window will take these two entry fields as arguments and run it through the speak_text() function, which we will learn to create in the next step.

audio_to_pdf()

This method, where we will convert the speech of an audio file to text in a PDF file, we will have two entry fields where the first one will ask for the path to an audio file that should only be in .wav and .mp3 formats and the second one will take the path to a PDF file.

speech_recognition()

The button in this window will take the two paths as arguments and run them through the speech_recognition() function where the audio file will be transcribed.

@staticmethod()

@staticmethod is used to convert a method indented under a Class definition to an autonomous function that is not bound to the scope of the Class.

We will take 2 arguments in the **speak_text()** method, which will be our path to the PDF file and the page number that the user wants the computer to read.

- At the start, we will check if the file path and the page number arguments are not empty. If they are, we will stop the function from further functioning.
- Then, we will initiate the pyttsx3 engine.
- After that, we will open our PDF file in the read mode, where we will get the page number the user asked for, extract the text from that page and then use the pyttsx3 engine to read the text.

DESIGNING

Design Patterns is the most essential part of Software Engineering, as they provide the general repeatable solution to a commonly occurring problem in software design. They usually represent some of the best practices adopted by experienced object-oriented software developers.

1. METHODOLOG :

System Design

There isn't enough time in people's lives these days to read a book or use internet services to convert a PDF file to an MP3 player. As a fellow bibliophile, I have a folder where I save all of the books I'd want to read but on no occasion get around to it. As a result, I reasoned, why not turn them into audiobooks so I could listen to them while doing other work? A Python GUI programme is being developed to convert the PDF file to audio format and play it back for the user on this machine .

No audio file or MP3 player is required for the application, making it easier for users to get started and utilise. After selecting a PDF file, the user can begin listening. In contrast to distinctive audiobook convertors, they convert PDF texts (or pictures) to speech and enable volume control for single-voice conversion (either male or female). When it comes to modifying the voice, the user has only one option. They provide the ability to play and pause the video. The pace of the voice is persistent



Reading stories or essays or any text can be arduous, however an audio reading of the text is convenient and doesn't require as much concentration as reading requires. In this project, I implemented a simple PDF to audio converter. This code scans page(s) of PDF and reads it using audio, to the user

2.PROJECT SCOPE

The translation of PDF text to audio signals is in high demand due to PDF's extensive use as the world's most popular document format. In addition to instructional purposes, they may be used for a variety of other tasks, such as automobile steering, train station announcements, telecommunications response services, and reading e-mail. PDF files cannot be viewed or read by those with vision damages, which is a big disadvantage. This study examines the problems of turning PDF text into spoken word. One of the ways to make PDF-based text seem more accurate when played over an audio system is to make synthetic speech sound more natural.

Literature Survey:

Before starting the project we did a survey on why people are interested in audiobooks rather than reading and also, we noted the reasons if they are not interested in audiobooks. Most of the interested people answers are that listening to a book is convenient than reading a book and it is very useful for the people who can understand the language but cannot read, so, through audiobook system they can listen to the books they liked even

though they cannot read. This Audiobook System is also useful for blind people so rather than reading they can listen to the book and understand or feel the book[4,5]. Some people those who don't like audiobook system the main reason is that they can improve their reading skills while reading a book rather than listening to a book so that they can be perfect in the language[6]. While some people say that they are happy that they don't need to carry a book every if they are suddenly interested in reading and some others say that they can do multitasking i.e., by listening to a book, they can do some other work at the same time. After reading the responses from the survey we got to know that there are more advantages than disadvantages of the audiobook system

3.Problem statement :

Because PDF is the most widely used document format on the planet, there is a demand for PDF text to be converted to audio signals. In addition to instructional objectives, they may be used for a variety of other tasks, such as auto mobile navigation, train station notifications, telecommunications reply services, and reading e-mail. PDF files are not accessible to those with poor vision, which is a severe drawback. In this research, the difficulties of translating PDF text to voice are examined.

In PDF-based writing, one challenge is to make the audio output of artificial speech sound more truthful. You'll notice when you look at our Audiobook converter that it converts PDF text (or pictures) into voice by making slight alterations inside the same PDF document. Voices may be converted. Using a single button press, the user may switch between male and female voices. Using voice speed as an example, it is possible to alter the pace at which one speaks (rapid ,normal, and slow).

Alternate methods of regulating the volume are available. Since the audiobook converts the PDF to text, we have complete control over what we want to include or exclude from the final product. Adding this additional competence to our audiobook converter is a first.

This makes it easier for the user to add or remove lines, for example. For example, the user may use this notepad to combine or delete sections of text as they usually would. While the music is playing, we may also open the relevant PDF or any other PDF.

ADVANTAGES OF THIS SYSTEM

- This application makes its users listening skills strong.
- The users will have ease in listening audiobooks.
- User-Friendly interface
- No prior environment required to run this application like Java, python.
- The query will be executed only in the command line interface and Graphical user Interface
- Free of cost to make this project.
- User can get text file as well as audiofile.

DISADVANTAGES OF THIS SYSTEM

- The user can select any of the two voices.
- The user has to kill the program in order to stop the audio file while running in GUI.
- Works fine only in Windows family, application is not compatible with macOS and Linux.

TESTING

TESTING

Let us see how to read a PDF that is converting a textual PDF file into audio.

Packages Used:

- **pyttsx3:** It is a Python library for Text to Speech. It has many functions which will help the machine to communicate with us. It will help the machine to speak to us
- **PyPDF2:** It will help to the text from the PDF. A Pure-Python library built as a PDF toolkit. It is capable of extracting document information, splitting documents page by page, merging documents page by page etc.

Both these modules need to be installed

```
pip install pyttsx3
```

```
pip install PyPDF2
```

```
# PDF to Audio Conversion in Python #
```

```
# Pip install has to be done in command prompt window by using the below comments
```

```
#pip install gTTS
```

```
#pip install PyPDF2
```

```
import PyPDF2 # Import the required module for text to speech conversion
```

```
from gtts import gTTS # This module is imported so that we can play the converted audio
```

```
import os
```

```
pdffile = open('F:\w2a.pdf', 'rb') # Input file path select the correct path
```

```
readfileip = PyPDF2.PdfFileReader(pdffile) # creating a pdf reader object
```

```
print(read file ip.numPages) # printing number of pages in pdf file
```

```
pageinfo = readfileip.getPage(0) # creating a page object

# extracting text from page

print(pageinfo.extractText()) # Printing the extrated text

text=pageinfo.extractText() # Assigning the exterated text to variable

pdffile.close() # closing the pdf file object

mytext = text # The text that you want to convert to audio

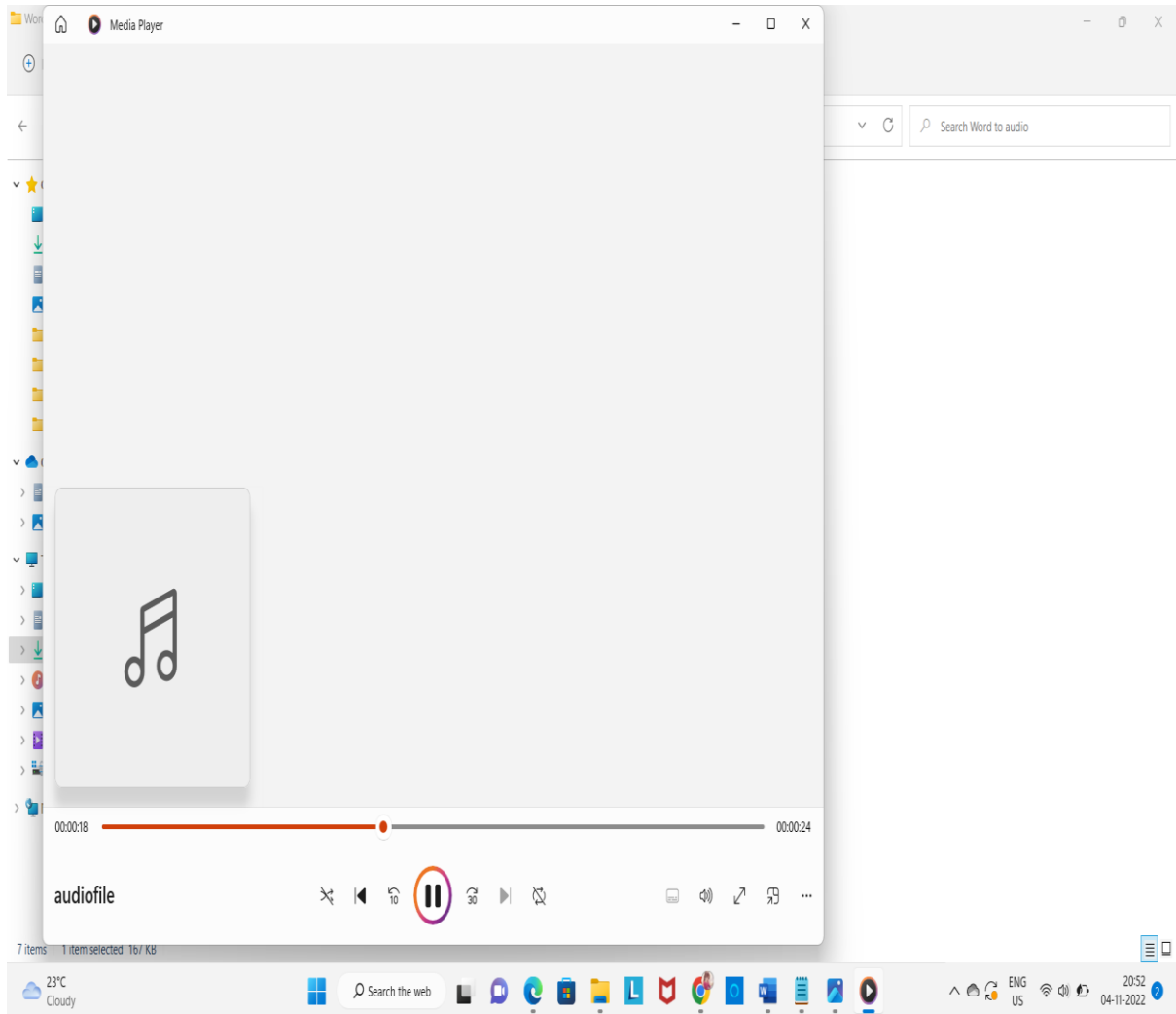
language = 'en' # Language in which you want to convert

# Passing the text and language to the sytem,here we have marked slow=False. Which tells
the module that the converted audio should have a high speed
audiofile = gTTS(text=mytext, lang=language, slow=False)

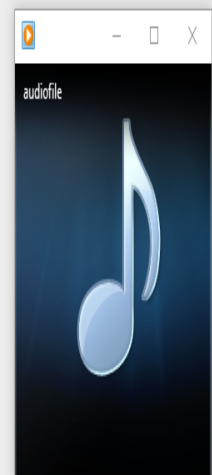
audiofile.save("audiofile.mp3") # Saving the converted audio in a mp3 file named Audio
file

os.system("audiofile.mp3") # Playing the converted audiofile.mp3
```

OUTPUT



Also, PEPS Tech offers a wide range of training in basic to highly advanced technologies. Also, we provide internships and projects with practical knowledge.



WORKFLOW:

1. When the user selects a pdf file, the pdf to audiobook converter begins the process of converting it into an audio book
2. After selecting a PDF file, we must type in the rate of speech per minute using the keyboard.
3. " In this way, we are able to set the amount of words per minute that we like to hear
4. Now we need to select the voice type, we can select male voice ore female voice. We can control the volume of the speech produced.
5. We can also extract .txt file using command prompt which helps in extracting the file in text format.
6. The exit button can be used to exit the application.

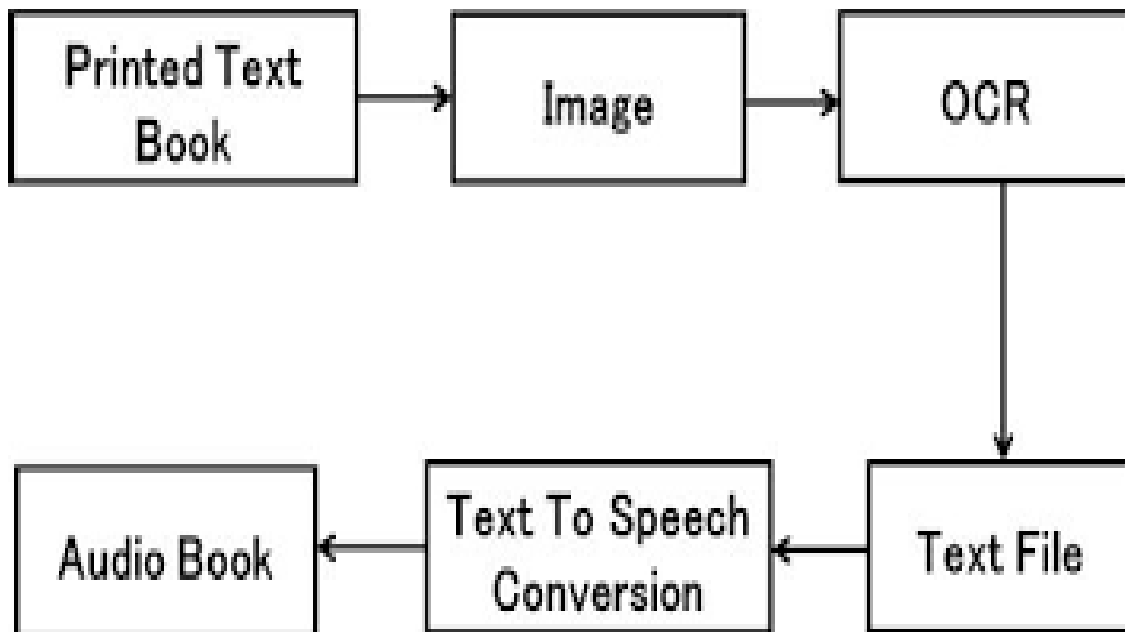
WORKING

In our project, we are converting a.pdf file into a.mp3 file as a core functionality.

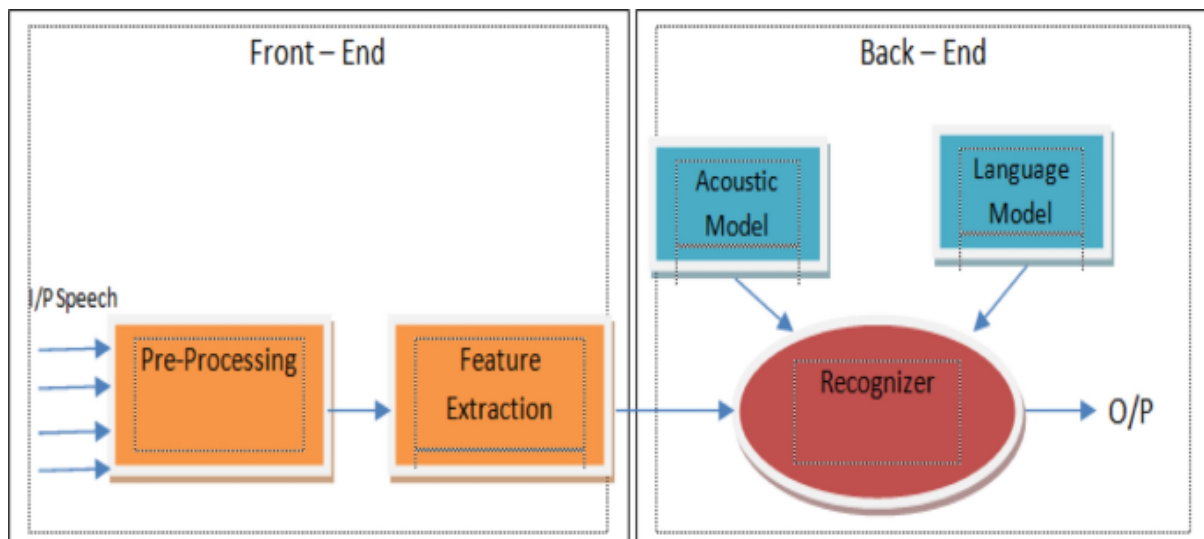
Two libraries, pyttsx3 and pdfminer, were used to accomplish this.

- One of the most popular programmes for mining data from PDF files is PDFMiner. It's the only PDF application that concentrates only on text data mining and analysis. If you use PDFMiner, you'll be able to see where text appears on a page, along with font and line information.
- A Python library for converting text to voice is available in the form of pyttsx3. If you don't have access to the Internet, this library can still be used. Extraction of data, page-by-page document splitting, merging, and more are all possible.
- The PyPDF2, pdfminer, and pyttsx3 modules will be imported.
- Using pdfminer, the text is then mined from the pdf file and saved as a text file.
- It will be transformed to an mp3 file using Google's text-to-speech module, gtts.
- Windows player will be used to play the output.

BLOCK DIAGRAM



FRONT-END AND BACK END



IMPLEMENTATION

4.IMPLEMANTATION:

4.1SOURCE CODE:

```
from tkinter import *

import tkinter.messagebox as mb
from path import Path
from PyPDF4.pdf import PdfFileReader as PDFreader, PdfFileWriter as PDFwriter
import pyttsx3
from speech_recognition import Recognizer, AudioFile
from pydub import AudioSegment
import os

class Window(Tk):
    def __init__(self):
        super(Window, self).__init__()
        self.title("ProjectGurukul PDF to Audio and Audio to PDF converter")
        self.geometry('400x250')
        self.resizable(0, 0)
        self.config(bg='Burlywood')
        Label(self, text='ProjectGurukul PDF to Audio and Audio to PDF converter',
              wraplength=400,
              bg='Burlywood', font=("Comic Sans MS", 15)).place(x=0, y=0)
        Button(self, text="Convert PDF to Audio", font=("Comic Sans MS", 15), bg='Tomato',
              command=self.pdf_to_audio, width=25).place(x=40, y=80)
        Button(self, text="Convert Audio to PDF", font=("Comic Sans MS", 15), bg='Tomato',
              command=self.audio_to_pdf, width=25).place(x=40, y=150)

    def pdf_to_audio(self):
        pta = Toplevel(self)
        pta.title('Convert PDF to Audio')
        pta.geometry('500x300')
        pta.resizable(0, 0)
        pta.config(bg='Chocolate')
        Label(pta, text='Convert PDF to Audio', font=('Comic Sans MS', 15),
              bg='Chocolate').place(relx=0.3, y=0)
        Label(pta, text='Enter the PDF file location (with extension): ', bg='Chocolate',
              font=("Verdana", 11)).place(x=10, y=60)
        filename = Entry(pta, width=32, font=('Verdana', 11))
```

```

filename.place(x=10, y=90)
Label(pta, text='Enter the page to read from the PDF (only one can be read): ',
bg='Chocolate', font=("Verdana", 11)).place(x=10, y=140)
page = Entry(pta, width=15, font=('Verdana', 11))
page.place(x=10, y=170)
Button(pta, text='Speak the text', font=('Gill Sans MT', 12), bg='Snow', width=20,
command=lambda: self.speak_text(filename.get(), page.get())).place(x=150, y=240)

```

```

def audio_to_pdf(self):
atp = Toplevel(self)
atp.title('Convert Audio to PDF')
atp.geometry('675x300')
atp.resizable(0, 0)
atp.config(bg='FireBrick')
Label(atp, text='Convert Audio to PDF', font=("Comic Sans MS", 15),
bg='FireBrick').place(relx=0.36, y=0)
Label(atp, text='Enter the Audio File location that you want to read [in .wav or .mp3
extensions only]:',
bg='FireBrick', font=('Verdana', 11)).place(x=20, y=60)
audiofile = Entry(atp, width=58, font=('Verdana', 11))
audiofile.place(x=20, y=90)
Label(atp, text='Enter the PDF File location that you want to save the text in (with
extension):', bg='FireBrick', font=('Verdana', 11)).place(x=20, y=140)
pdffile = Entry(atp, width=58, font=('Verdana', 11))
pdffile.place(x=20, y=170)
Button(atp, text='Create PDF', bg='Snow', font=('Gill Sans MT', 12), width=20,
command=lambda: self.speech_recognition(audiofile.get(), pdffile.get())).place(x=247,
y=230)

```

```

@staticmethod

```

```

def speak_text(filename, page):
if not filename or not page:
mb.showerror('Missing field!', 'Please check your responses, because one of the fields is
missing')
return

```

```

reader = PDFReader(filename)
engine = pyttsx3.init()
with Path(filename).open('rb'):
page_to_read = reader.getPage(int(page)-1)
text = page_to_read.extractText()
engine.say(text)
engine.runAndWait()

```

```

@staticmethod
def write_text(filename, text):
    writer = PDFwriter()
    writer.addBlankPage(72, 72)
    pdf_path = Path(filename)
    with pdf_path.open('ab') as output_file:
        writer.write(output_file)
        output_file.write(text)
def speech_recognition(self, audio, pdf):
    if not audio or not pdf:
        mb.showerror('Missing field!', 'Please check your responses, because one of the fields is missing')

```

return

```

audio_file_name = os.path.basename(audio).split('.')[0]
audio_file_extension = os.path.basename(audio).split('.')[1]
if audio_file_extension != 'wav' and audio_file_extension != 'mp3':
    mb.showerror('Error!', 'The format of the audio file should only be either "wav" and "mp3"!')
if audio_file_extension == 'mp3':
    audio_file = AudioSegment.from_file(Path(audio), format='mp3')
    audio_file.export(f'transcript.wav', format='wav')
    source_file = 'transcript.wav'
    r = Recognizer()
    with AudioFile(source_file) as source:
        r.pause_threshold = 5
        speech = r.record(source)
        text = r.recognize_google(speech)
        self.write_text(pdf, text)

```

```

from tkinter import *
import tkinter.messagebox as mb
from path import Path
from PyPDF4.pdf import PdfFileReader as PDFreader, PdfFileWriter as PDFwriter
import pyttsx3
from speech_recognition import Recognizer, AudioFile
from pydub import AudioSegment
import os

```

Initializing the GUI window

```

class Window(Tk):
def __init__(self):
    super(Window, self).__init__()

```

```

self.title("ProjectGurukul PDF to Audio and Audio to PDF converter")
self.geometry('400x250')
self.resizable(0, 0)
self.config(bg='Burlywood')
Label(self, text='ProjectGurukul PDF to Audio and Audio to PDF converter',
wraplength=400,
bg='Burlywood', font=("Comic Sans MS", 15)).place(x=0, y=0)
Button(self, text="Convert PDF to Audio", font=("Comic Sans MS", 15), bg='Tomato',
command=self.pdf_to_audio, width=25).place(x=40, y=80)
Button(self, text="Convert Audio to PDF", font=("Comic Sans MS", 15), bg='Tomato',
command=self.audio_to_pdf, width=25).place(x=40, y=150)
def pdf_to_audio(self):
pta = Toplevel(self)
pta.title('Convert PDF to Audio')
pta.geometry('500x300')
pta.resizable(0, 0)
pta.config(bg='Chocolate')
Label(pta, text='Convert PDF to Audio', font=('Comic Sans MS', 15),
bg='Chocolate').place(relx=0.3, y=0)
Label(pta, text='Enter the PDF file location (with extension): ', bg='Chocolate',
font=("Verdana", 11)).place(x=10, y=60)
filename = Entry(pta, width=32, font=('Verdana', 11))
filename.place(x=10, y=90)
Label(pta, text='Enter the page to read from the PDF (only one can be read): ',
bg='Chocolate',
font=("Verdana", 11)).place(x=10, y=140)
page = Entry(pta, width=15, font=('Verdana', 11))
page.place(x=10, y=170)
Button(pta, text='Speak the text', font=('Gill Sans MT', 12), bg='Snow', width=20,
command=lambda: self.speak_text(filename.get(), page.get())).place(x=150, y=240)
def audio_to_pdf(self):
atp = Toplevel(self)
atp.title('Convert Audio to PDF')
atp.geometry('675x300')
atp.resizable(0, 0)
atp.config(bg='FireBrick')
Label(atp, text='Convert Audio to PDF', font=("Comic Sans MS", 15),
bg='FireBrick').place(relx=0.36, y=0)
Label(atp, text='Enter the Audio File location that you want to read [in .wav or .mp3
extensions only]:',
bg='FireBrick', font=('Verdana', 11)).place(x=20, y=60)
audiofile = Entry(atp, width=58, font=('Verdana', 11))
audiofile.place(x=20, y=90)

```



```

Label(atp, text='Enter the PDF File location that you want to save the text in (with
extension):', bg='FireBrick', font=('Verdana', 11)).place(x=20, y=140)
pdffile = Entry(atp, width=58, font=('Verdana', 11))
pdffile.place(x=20, y=170)
Button(atp, text='Create PDF', bg='Snow', font=('Gill Sans MT', 12), width=20,
command=lambda: self.speech_recognition(audiofile.get(), pdffile.get())).place(x=247,
y=230)
@staticmethod
def speak_text(filename, page):
if not filename or not page:
mb.showerror('Missing field!', 'Please check your responses, because one of the fields is
missing')

return
reader = PDFReader(filename)
engine = pyttsx3.init()
with Path(filename).open('rb'):
page_to_read = reader.getPage(int(page)-1)
text = page_to_read.extractText()
engine.say(text)
engine.runAndWait()

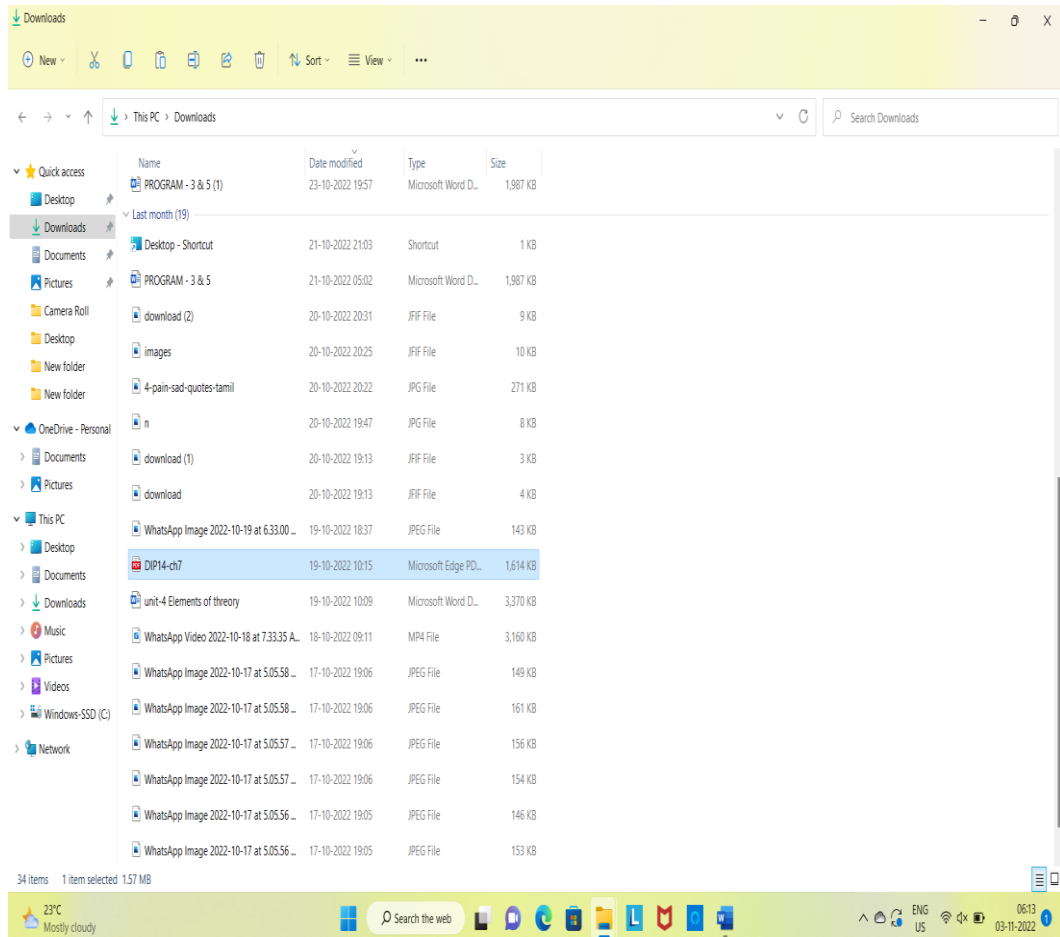
@staticmethod
def write_text(filename, text):
writer = PDFWriter()
writer.addBlankPage(72, 72)
pdf_path = Path(filename)
with pdf_path.open('ab') as output_file:
writer.write(output_file)
output_file.write(text)
def speech_recognition(self, audio, pdf):
if not audio or not pdf:
mb.showerror('Missing field!', 'Please check your responses, because one of the fields is
missing')
return

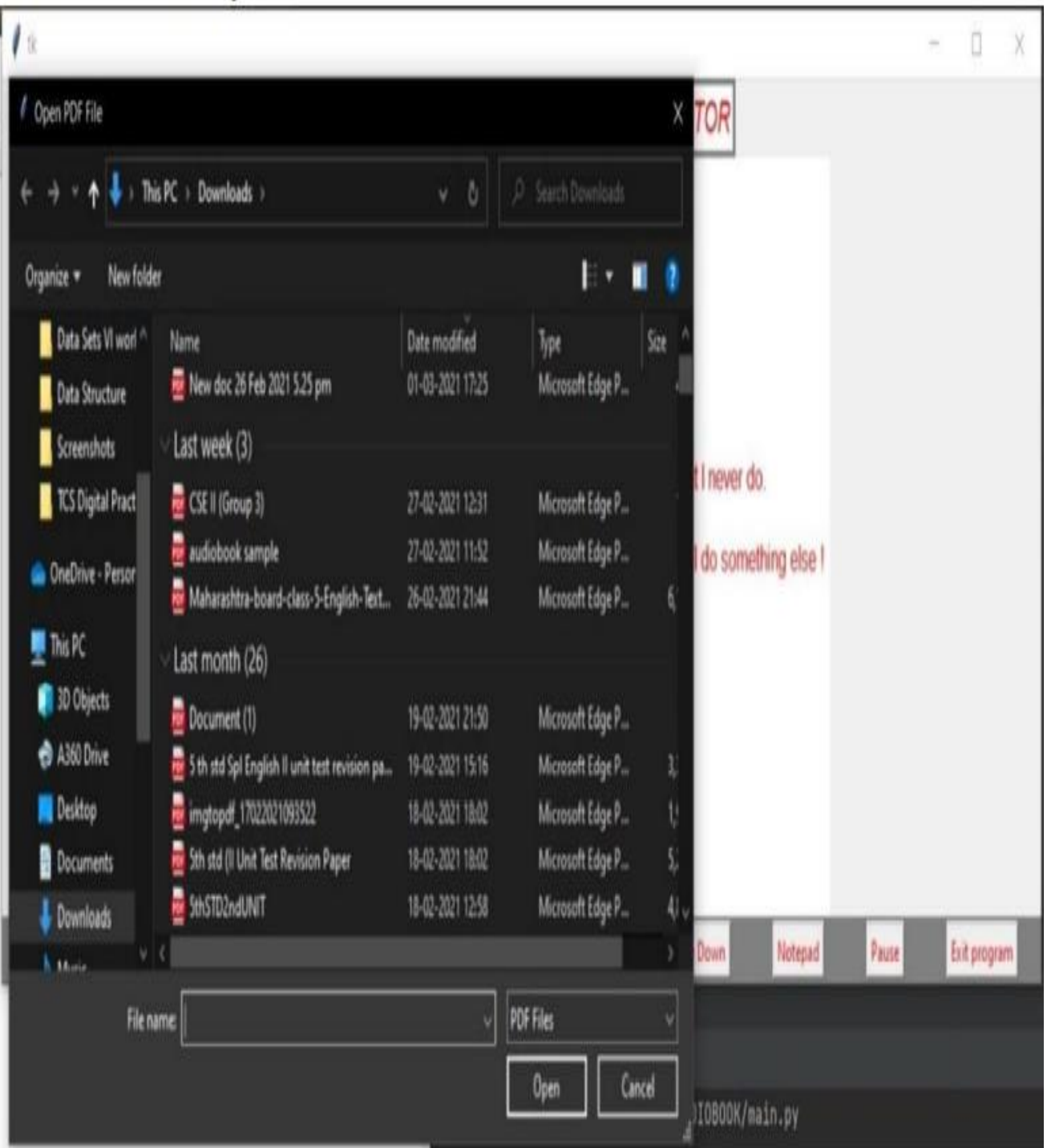
audio_file_name = os.path.basename(audio).split('.')[0]
audio_file_extension = os.path.basename(audio).split('.')[1]
if audio_file_extension != 'wav' and audio_file_extension != 'mp3':
mb.showerror('Error!', 'The format of the audio file should only be either "wav" and "mp3"!')
if audio_file_extension == 'mp3':
audio_file = AudioSegment.from_file(Path(audio), format='mp3')
audio_file.export(f'{audio_file_name}.wav', format='wav')

```

```
source_file = f'{audio_file_name}.wav'
r = Recognizer()
with AudioFile(source_file) as source:
    r.pause_threshold = 5
    speech = r.record(source)
    text = r.recognize_google(speech)
    self.write_text(pdf, text)
# Finalizing the GUI window
app = Window()
app.update()
app.mainloop()
```

4.2 SAMPLE INPUT:





SAMPLE OUTPUT:





ProjectGurukul PDF to Audio and Audio to...



ProjectGurukul PDF to Audio and Audio to PDF converter

Convert PDF to Audio

Convert Audio to PDF

Files and Consoles15self.title("ProjectGurukul")

Convert Audio to PDF

Convert Audio to PDF

Enter the Audio File location that you want to read [in .wav or .mp3 extensions only]:

Enter the PDF File location that you want to save the text in (with extension):

Create PDF

ProjectGurukul

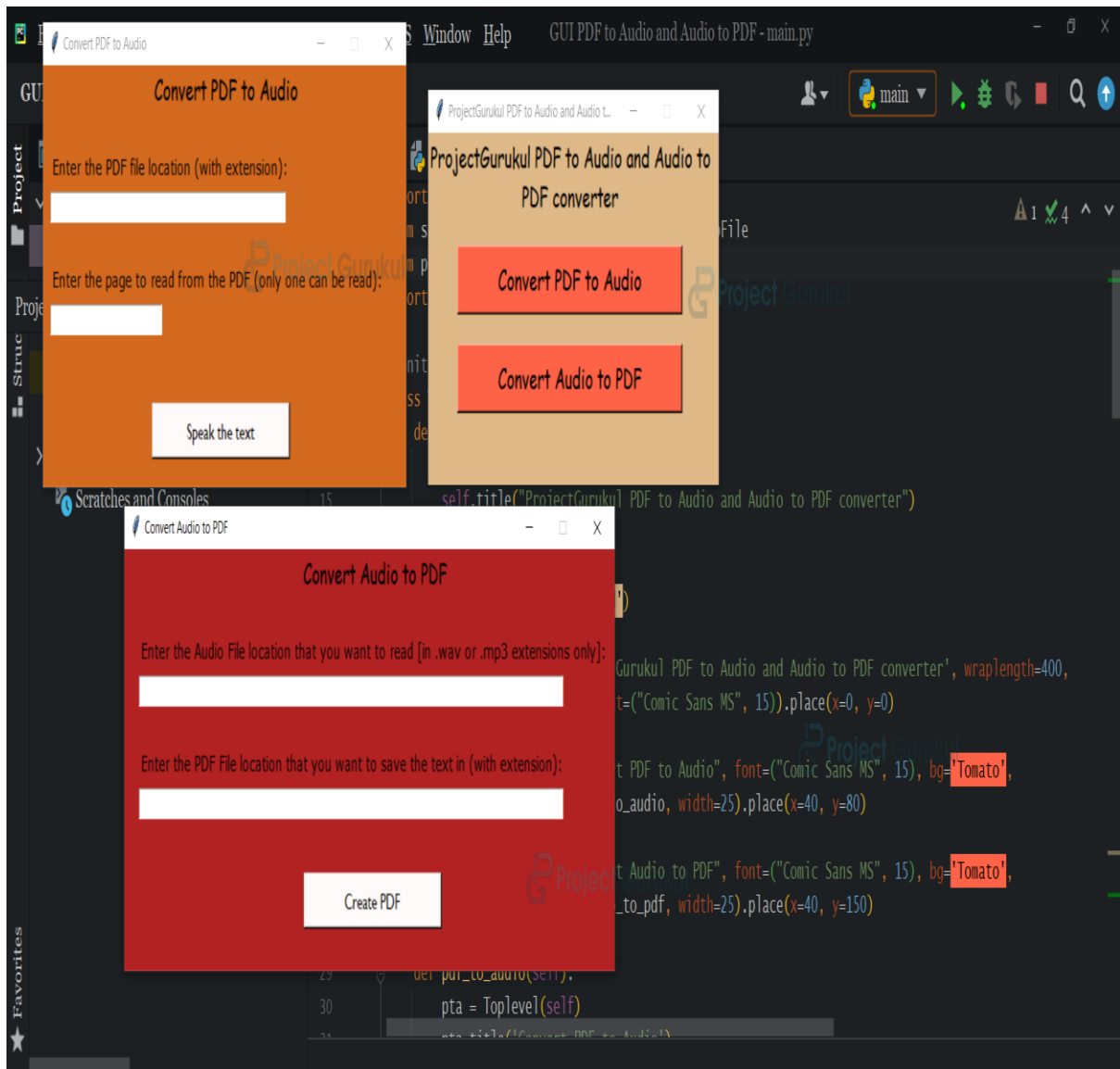
Future Scope:

As per our project present, we declare that a person who is having the interest to learn English if he comes from his native language background our project helps then more present we are having 4 modules in that like speech-text conversion, text-speech conversion, pdf-speech conversion, image-pdf conversion this concept in a single platform ad's beauty to our project.

In further we want to update this project by adding some more modules which can help future generation people access a variety of functions related to the library in this platform that makes the library system of student into easy way as per the present technology this type of conversions have more impact on the generation as we can take an example of Alexa and so on which makes the life in a smooth way.

In the same manner, the conversions also play a major role. In the future, we will implement our project in life by having a library in this virtual manner where we can upload any book into it and get the content of the book as a voice where just by listing, we can just gain knowledge in the best way as the every one say the thing which we hear makes a major impact.

FINAL OUTPUT:



CONCLUSION

Conclusion:

This mini project developed to easy to read the pdf file.It was seen that this code performs really well in reading straightforward PDF text files.

Should enable users to select the desired PDF and convert it to audio and display text in, so the user can understand that particular text has been read. Should enable students with reading disabilities.The success of this research project is significant given the broad use of audiobooks in literacy and library programs across the United States.

Teachers and school librarians may also use these findings as a rationale for adding audiobooks to the list of reading strategies used successfully with struggling readers.

We are interested in future research on the use of audiobooks with struggling readers who are younger and older than those who participated in the study and on audiobook usage with At this point, the code does not have a stop feature, I intend to add those and do more interesting things with the application of Machine Learning in the audiobook.

With the help of machine learning, we can add the features that will recognize the voice of the user and implement the function as the user wants . This feature will help mostly for the disabled users like the blind, handicapEnglish Language Learners .